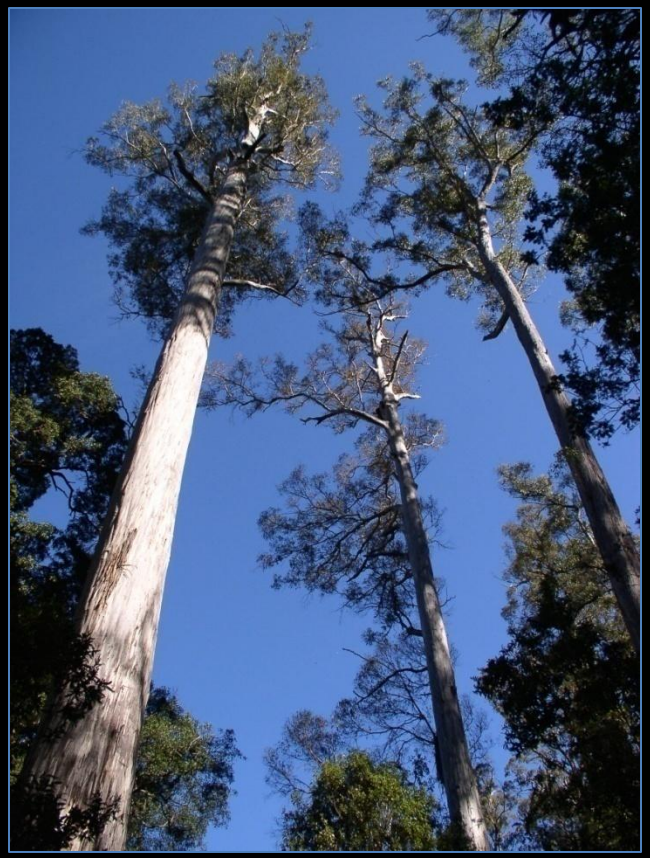
Measurement of Tree Basal Area & Volume





FOR 1001

Dr. Thom Erdle

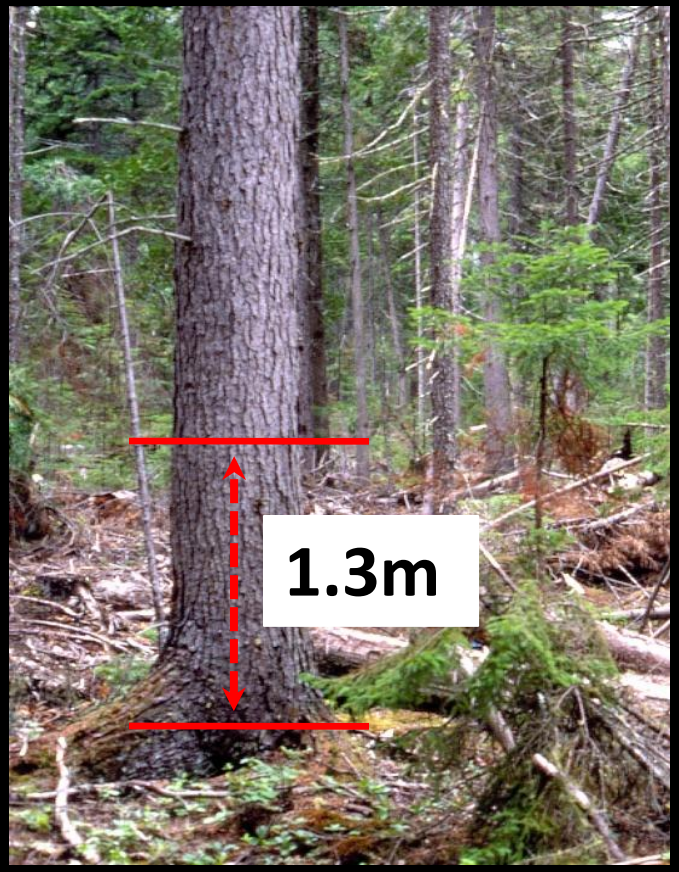
Today's Objectives

- Tree → Stand → Forest
- Basic geometric calculations
- Measurement of tree diameter
- Measurement of tree height
- Determination of:
  - basal area
  - volume
  - biomass & carbon
  - $ value

Tree Basal Area

What?

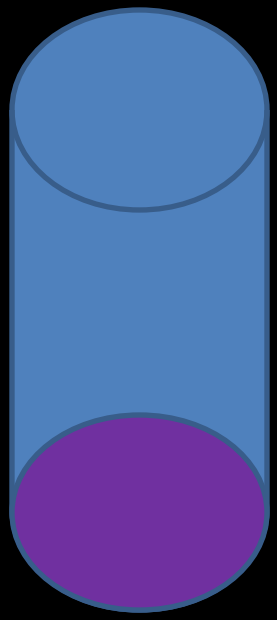
- **Simply the *cross-sectional area* of a tree stem measured at *breast height* (1.4m)**





Tree Basal Area

Why?



- **Basal area closely relates to *volume***
- ***Easy to* measure & calculate**
- **Useful measure of *site occupancy***
- **Useful to *forecast* future development of stand**

Tree Basal Area

How?

- **Area of a circle**

$$\text{Area} = \text{pi} * (\text{radius})^2$$

- **Diameter is twice the radius**

$$\text{Radius} = \text{diameter} / 2$$

- **Area of circle in terms of diameter is**

$$\text{area} = \text{pi} * (\text{diameter}/2)^2$$

$$\text{area} = \text{pi} * \text{diameter}^2 / 4$$

Tree Basal Area

How?

- Convert diameter of a circle to area of a circle

$$\text{area} = \text{pi} * \text{diameter}^2 / 4$$

$$\text{Basal area} = \text{pi} * (\text{DBH})^2 / 4$$



- But DBH is usually in *cm* & basal area is usually expressed in *square metres*
- 10000 *cm* per *square metre*

$$\text{Basal area} = \text{pi} * (\text{DBH})^2 / (4*10000)$$

$$\text{Basal area} = \text{pi} * \text{DBH}^2 / 40000$$

Where: basal area in in m^2
DBH is in cm

Tree Basal Area

How?

- **Forestry naming & unit convention**

Basal area denoted by "BA"

Basal area = in m^2

Diameter at breast height = DBH in cm

- **Final form for basal area calculation**

$$BA_{(m^2)} = pi * DBH_{(cm)}^2 / 40000$$

Tree Volume Calculation

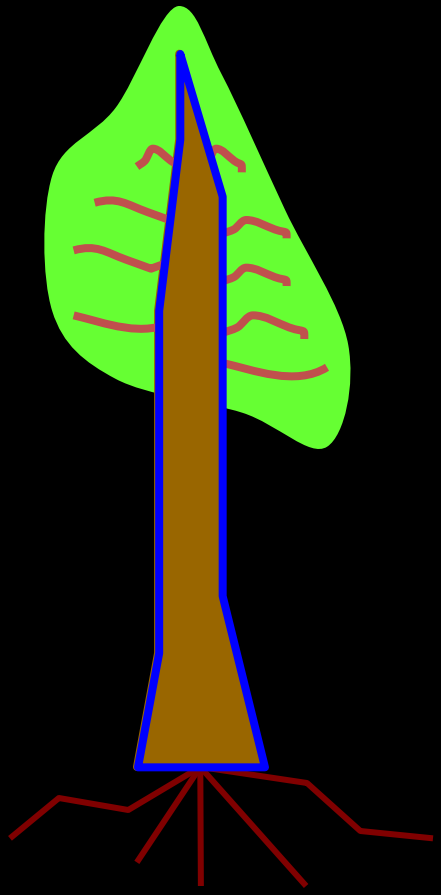


What?

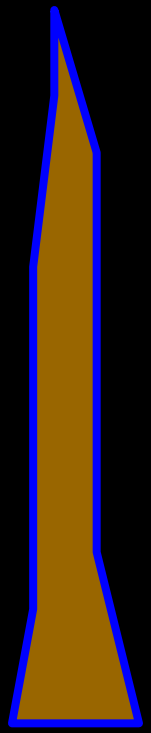
- Volumetric content of tree
- Volume of different portions of tree (know which one you are talking about)
- ***Total*** volume (main stem from ground to tip)
- ***Merchantable*** volume (main stem excluding stump and tip defined to a minimum diameter)

Tree Volume Calculation

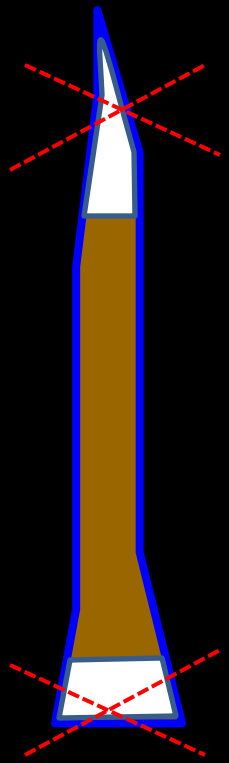
What?



Total volume



Merchantable volume



Tip

Stump

Tree Volume Calculation

Why?

- ***Product content*** **and tree** ***value*** **are directly related to tree volume**
- ***Carbon*** **and** ***biomass*** **are directly related to tree volume**

Tree Volume Calculation

How?

- Water displacement
- Geometry of solid shape approximating tree stem shape
- Cut stem into sections, measure & sum



Either physically cut or measure sections on uncut stem

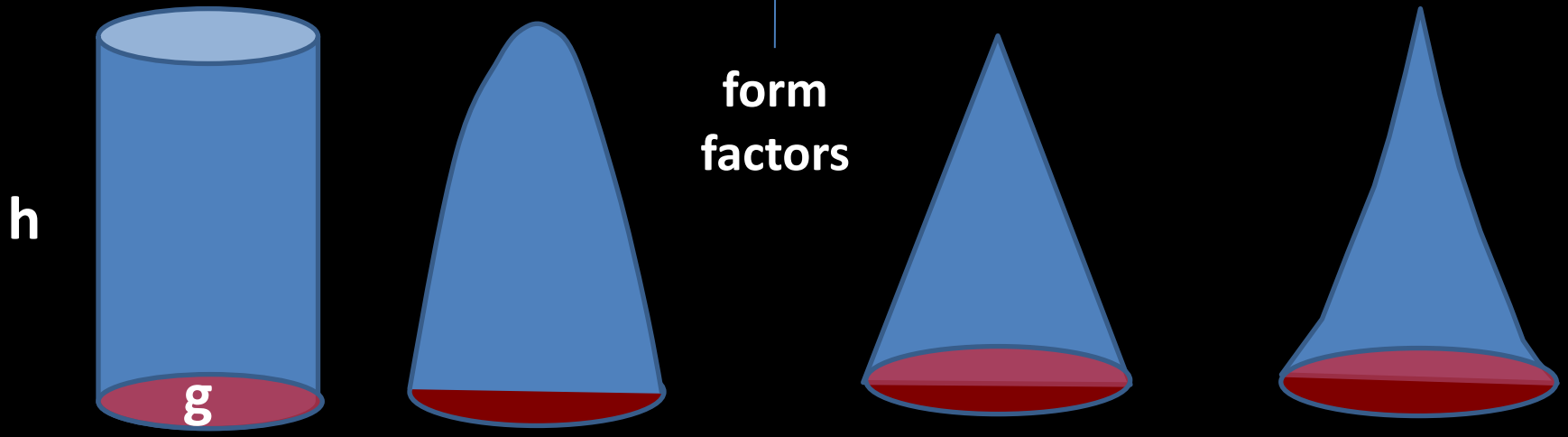
Tree Volume Calculation

How?

- **Geometry of solid shape approximating tree stem shape**

g = cross sectional area of base; h = height; v = volume

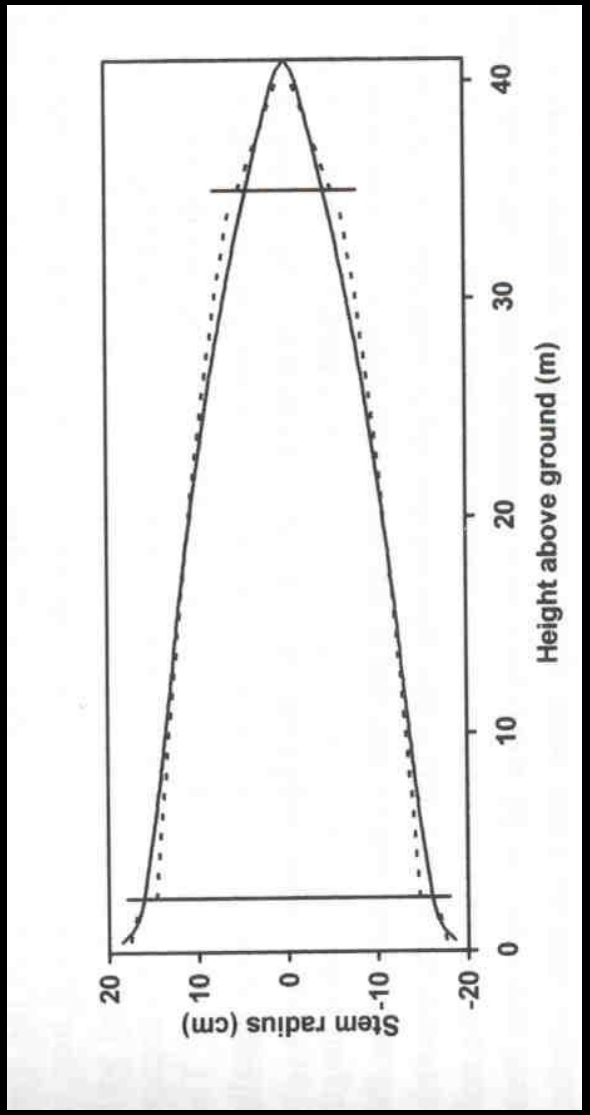
	cylinder	parabola	cone	neloid
V =	1 * g * h	0.5 * g * h	0.33 * g * h	0.25 * g * h



Tree Volume Calculation

How?

- **Tree form a mix of all these shapes**
- **Shape changes along tree stem**
- **So we calculate the form factor using stem analysis**



Tree Volume Calculation

How?

- Section tree (either cut or mark)
- Measure each section to calculate volume



- Sum all sections to obtain tree total
- Perform for many trees across range of sizes

Tree Volume Calculation

How?

- Relationship varies somewhat by tree, region and treatment
- E.g. for Noonan (Kershaw)

$$V = 0.42 * BA * H$$

- Where: V = tree volume (m3)
  BA = tree basal area (m2)
  H = tree height (m)

Called *Volume Equations*

- Sometimes more complex equations are developed

$$V = D^a * [H^2/(H-1.4)]^b * e^c$$

Tree Volume Calculation

How?

- **Sometimes tables are constructed that list volume by DBH & height**

Called *Volume Tables*

DBH	Spruce		Fir	
Class	Height (m)	Volume (m^3)	Height (m)	Volume (m^3)
2	2.2	0.000	2.0	0.000
4	4.2	0.003	3.9	0.003
6	6.0	0.009	5.6	0.008
8	7.6	0.020	7.2	0.019
10	9.1	0.037	8.6	0.034
12	10.5	0.060	10.0	0.055
14	11.8	0.090	11.2	0.082
16	12.9	0.128	12.4	0.116
18	14.0	0.173	13.4	0.156
20	14.9	0.226	14.4	0.203
22	15.8	0.286	15.3	0.257
24	16.6	0.355	16.1	0.318

Tree Volume Calculation

Know

- **Volume equations allow you to calculate tree *volume* from tree *diameter* and *height***
- **They can be *complex* and very *accurate***
- **Rough-and-ready *approximation***

$$V = 0.42 * g * h$$

- ***Volume tables* allow you to *"look up"* tree volume from tree diameter and height**